

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 4

Cryogenics

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1. HELIUM LEVEL INDICATOR (HLI) WITH THE RMMU:

To be able to measure the helium level inside the F2000 type magnet the RMMU has to be installed. Note that the RMMU can be used stand alone. Stand alone use of the RMMU means that the fibers that are connected to the computer are not needed.

NOTE

The Helium Level Indicator (HLI mounted inside the Remote Magnet Monitor Unit, RMMU) allows the operator to monitor the level of liquid helium inside the helium vessel. The HLI is an indicator only and is intended to give an approximate measurement of the liquid helium level. The inaccuracy is $\pm 5\%$ of the full scale which equals 60 litres !

Procedure to measure the helium level with using a Digital Volt Meter (DVM):

1. If not yet done, connect the RMMU to the mains and to the magnet via the instrumentation cables.
- MRJ4 at the RMMU to MBJ1 at the magnet.
2. Connect a DVM, (set to 100 mV), to the voltage jacks on the RMMU
3. Press the HLI button.
4. Read the DVM after 30 seconds; reading in mV equals '%'. Write down the value, date and time in the MRL.

Note that when the system is installed the following procedure can be used:

Procedure to measure the Helium Level using the Gyroscan system:

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: READ HELIUM LEVEL.
Helium level is ..%

2. CHECKING IF LIQUID HELIUM IS INSIDE THE MAGNET.

Before starting to fill the magnet the temperature inside the helium vessel must be known. Three situations are possible:

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1. The helium vessel contains more than 0 litres of liquid helium.
Filling can be done with the refrigerator installed and running.
If the refrigerator is not installed yet, install it as soon as possible.
In this case the 'normal' helium filling procedure must be followed.
2. The helium vessel is empty and the temperature is lower than 150 K (*).
Start filling helium as soon as possible. Use the bottom fill procedure. See paragraph 4.2.
3. The helium vessel is empty and the temperature is higher than 150 K (*).
In this case the magnet needs to be precooled with liquid nitrogen.
For this action you have to ask assistance at the MR Helpdesk.

(*) See the following paragraph for determining the temperature inside the helium vessel.

3. DETERMINING THE TEMPERATURE INSIDE F2000 MAGNETS

3.1. OBJECTIVE

The objective of this procedure is to determine the temperature inside the magnet vessel when the Helium Level Indicator shows that the magnet is empty. This procedure is only applicable to F2000 magnets. If the temperature inside the helium vessel is lower than 150 K, filling directly with liquid helium is possible. If the temperature inside the helium vessel is higher than 150 K, precooling with liquid nitrogen is recommended. The decision to start precooling with liquid nitrogen at 150 K and not at a different temperature, is an economical one. Precooling with liquid nitrogen may only be performed by qualified engineers.

3.2. REQUIRED EQUIPMENT AND DOCUMENTS

- Retractable current leads (+) and (-).
- 10 V power supply (laboratory type, capable of supplying > 1 mA).
- Digital Volt Meter (input impedance > 1 MOhm for DC volts range).
- The 'Electrical Interconnection Drawing 3001' delivered with each magnet.
This document is helpful for understanding the measurement, but is not strictly necessary.

3.3. PRECONDITIONS

The measurement can be carried out on the bare magnet, without cabling and without magnet power supply cabinet. Installation of the magnet inside the RF shielded room is not necessary. Never perform this measurement on a magnet on field! If the magnet is on field, its temperature is always 4.2 K.

3.4. PROCEDURE

1. Make sure that the magnet is ramped down.
2. Insert both current leads into the magnet.
3. Measure the resistance between the positive and negative current lead. Do not measure at the jacks on the current leads, but measure on the big connectors at the end of the current leads.
If the resistance is smaller than 0.5 Ohm, the magnet winding and/or the magnet main heater are still superconductive, meaning that the temperature inside the magnet is lower than 10 K. If the resistance is

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higher than 0.5 Ohm proceed with step 4. In the latter case the measured resistance is unreliable due to the presence of the magnet main heater.

4. Disconnect the instrumentation cable FM1 from the magnet.
5. Connect the output of the 10 V power supply between the big connector of the (-) current lead and pin J of the 41-pin connector MJ1 on the magnet see Figure 1 and Figure 2. A current of about 0.5 mA should flow.
6. Measure the voltage U_{cm} between pins C and M of the 41 pin connector MJ1 on the magnet.
7. Calculate the magnet resistance R_m as follows.

$$R_m = \frac{U_{cm} \text{ (Volt)}}{0.5 \text{ (mA)}} \text{ [Ohm]}$$

8. Determine the approximate temperature of the coils from the Table 1.
9. Remove the 10 V power supply and reinstall the instrumentation cable FM1.

Figure 1 - Magnet temperature measuring set up

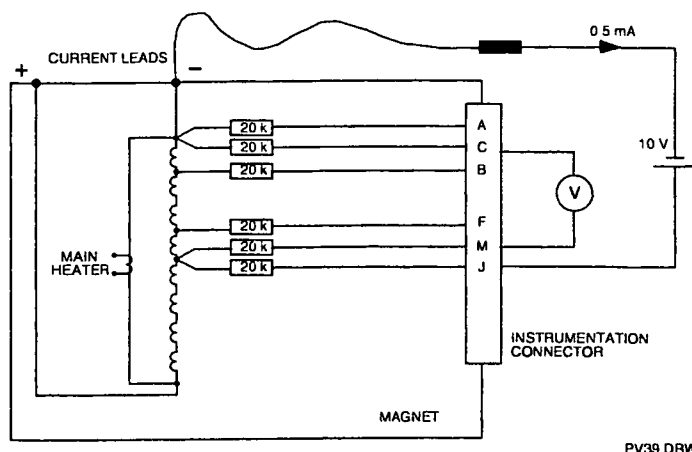
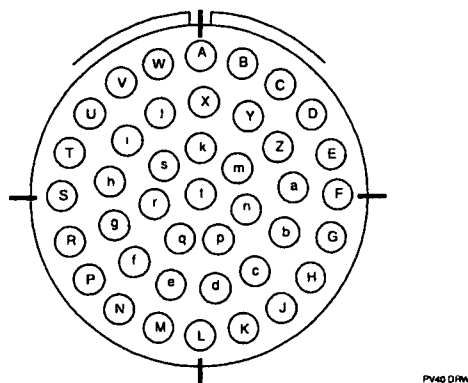


Figure 2 - Instrumentation connector



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Table 1 - Conversion table from resistance to temperature for F2000 magnets.

TEMPERATURE	F2000 0.5 TESLA		F2000 1.0 TESLA		F2000 1.5 TESLA	
300 K	53	OHM	125	OHM	200	OHM
200 K	37	OHM	81	OHM	142	OHM
150 K	24	OHM	55	OHM	90	OHM
100 K	10	OHM	24	OHM	38	OHM
50 K	2.0	OHM	5	OHM	9	OHM
10 K	0.4	OHM	1.0	OHM	1.7	OHM

NOTE

When the refrigerator system is working and the magnet is running out of liquid helium, leave the refrigerator system on and running!

4. LIQUID HELIUM FILLING**4.1. ONLY QUALIFIED ENGINEERS**

Filling helium on field is only allowed for qualified engineers. The service engineer can be considered 'qualified' when he/she has participated in a training which included the helium filling process, safety aspects and experienced one filling action. With 'experienced' we mean actually inserting the transfer line inside the dewar / magnet and being familiar with the 'whistling' sound produced by a transfer line when the dewar is almost empty. Just being present with a fill action is not enough.

NOTE

If 'warm' helium gas has entered a ramped down magnet during helium filling, it is not allowed to ramp the magnet within the first 12 hours. If the magnet is ramped up before this time has passed, a magnet quench may be the result! It can take up to 12 hours before 'warm' gas has cooled down enough inside the magnet to allow for ramping up again.

4.2. INTRODUCTION AND SAFETY

Liquid helium must be maintained inside the helium vessel to keep the windings super conductive. Liquid helium level probes are installed inside the helium vessel to monitor the level of liquid helium. The probes are connected to the Helium Level Indicator (HLI).

In general, the helium level is monitored every 24 hours to make sure that the level of liquid helium is adequate. For a magnet that is on field this level should be higher than 30%. Under normal circumstances the liquid helium consumption will be approximately 1 % / week.

The following safety precautions must be observed when filling a magnet with liquid helium:

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- Always store and transfer liquid helium under positive pressure and in closed systems to prevent contamination of the cryogen by air.
- Do not permit condensed air on transfer tubes to run down into the magnet or dewar opening as it may freeze up and block the release of helium vapors.
- Keep pressure relief devices free and clear.

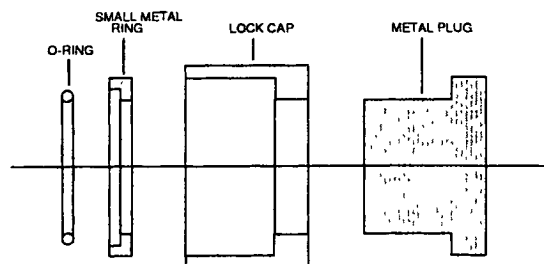
WARNING

- *During helium filling, it is important to have room ventilation for escaping helium gas. For mobile systems the hatch must be opened to ensure good ventilation.*
 - *Helium gas supply bottles must be kept outside the 5 gauss line.*
- *Liquid helium is extremely cold. The liquid, or cold vapor from the liquid, can cause cold contact burns and damage to the eyes and skin. Do not touch vapor plumes or frosted pipes or valves. Protective equipment and clothing approved for use with cryogenics must be worn to protect the eyes and skin whenever handling or working with the liquid helium or cold vapor from the liquid. For details about helium see the Material Safety Data Sheet MSDS MR-005.*

CAUTION

1. *Check for leaks several hours after helium fill has finished. After this time the helium should have built up sufficient pressure to make it possible to detect leaks.*
 2. *Cold/frozen O-rings should NEVER be mechanically loaded. The O-rings have to be at room temperature before the compression fittings of the ports are tightened.*
 3. *Be sure that the helium fill opening is provided with the following parts and that a spare set is available:*
 - O-ring 4522 130 73601
 - Small metal ring 4522 130 73611
 - lock cap 4522 130 73621
 - Metal plug 4522 150 10801
- If one of these items is not present, the seal is not leak tight. See Figure 3.*
4. *Do not leave the magnet open for a long time. If the magnet does not have over pressure, air can enter the helium vessel where it will freeze solid and cause icing problems.*

Figure 3 - Compression fitting



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Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
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The liquid helium filling procedure described on the following pages, should be used when there is still some liquid helium inside the magnet.

If no liquid helium is inside the magnet and the temperature inside the magnet is lower than 150 K, the 'bottom fill procedure' should be used.

Bottom fill procedure:

At point 15 of the liquid helium fill procedure, the instruction 'Transfer line is inserted 40 cm into the magnet to prevent a bottom fill' should be changed into:

Insert the transfer line all the way down into the bottom fill funnel. By wiggling the transfer line one can determine if it is seated properly in the bottom fill funnel.

4.3. TOOLS NEEDED FOR FILLING HELIUM

Tools needed:

- Liquid helium transfer line (not delivered with the system). See paragraph: 4.3.1
- High-pressure cylinder of helium gas or self pressurizing dewar.
- Non-magnetic liquid helium dewars.
- Pressure regulator for helium gas with fittings and hose. Pressure regulator not needed if self-pressurized dewars are used. Fittings depending on the dewar configuration. With the pressure regulator it should be possible to adjust the pressure between 0 and 10 psi / 700 mBar.
- Protective gloves and face shield.
- Aluminum step ladder, height about 180-200 cm.
- Snoop, for checks of seals. Delivered with the system.
- Heat gun of minimal 1500 Watt.
- Spare compression fittings

4.3.1. LIQUID HELIUM TRANSFER LINES

The position of the transfer line input of a F2000 magnet is located higher than on previous magnet types. Therefore the height of the transfer line is important. Table 2 shows transfer lines that can be used, provided that the ceiling height is sufficient.

Philips 12 nc	Vendor p/n	Vendor	Min c. height NT	Min c. height F2000	Figure:
4522 300 02892	621.0635	OMT	2716	3080	Figure 4
4522 150 00143	30114-01	IGC-T5/NT	2620	3186	Figure 5
4522 150 23101	n.a	Air products	2555	2885	Figure 6
4522 150 23101+23481	n.a	Air products	--	2650	Figure 6
Not available	On request	Linde L-form	2565	3085	
Not available	On request	Linde -dip tube	Inserting outside		
Not available	3 221 3451	Linde U-form	2735	2915	
4522 150 23611	K0007	IGC-F2000	2620	2620	Figure 7

Table 2: Helium transfer lines

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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Figure 4 - OMT Helium transfer line

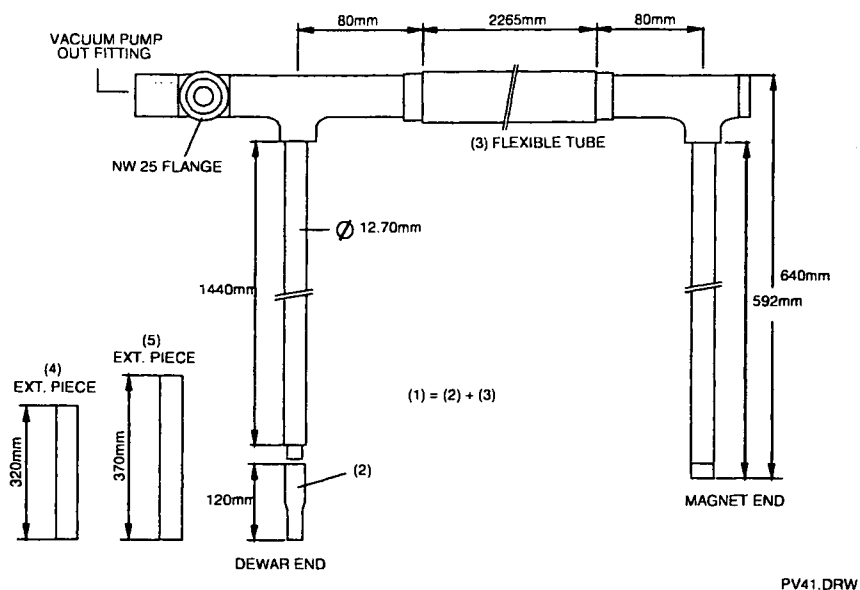
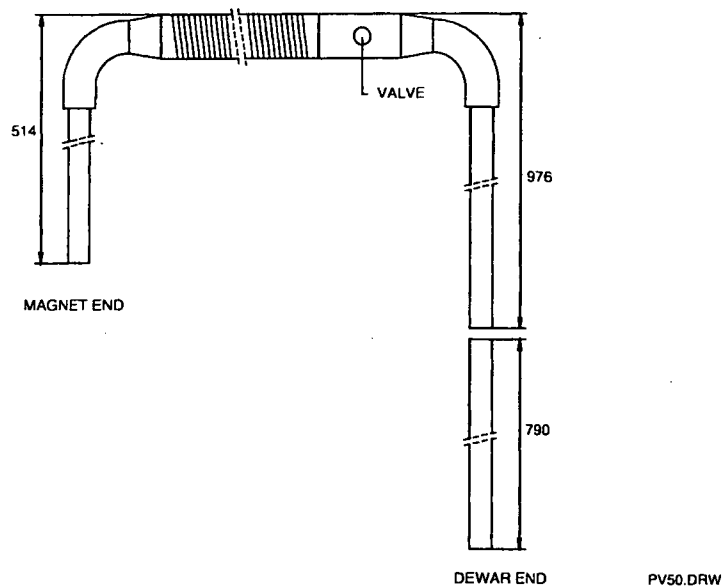


Figure 5 - IGC Helium transfer line



Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Figure 6 - Air Products helium transfer line

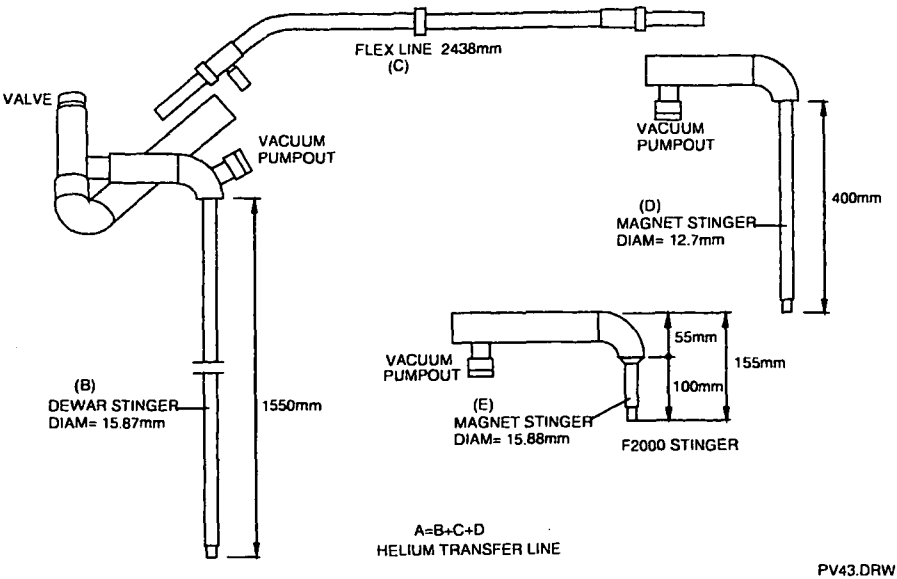
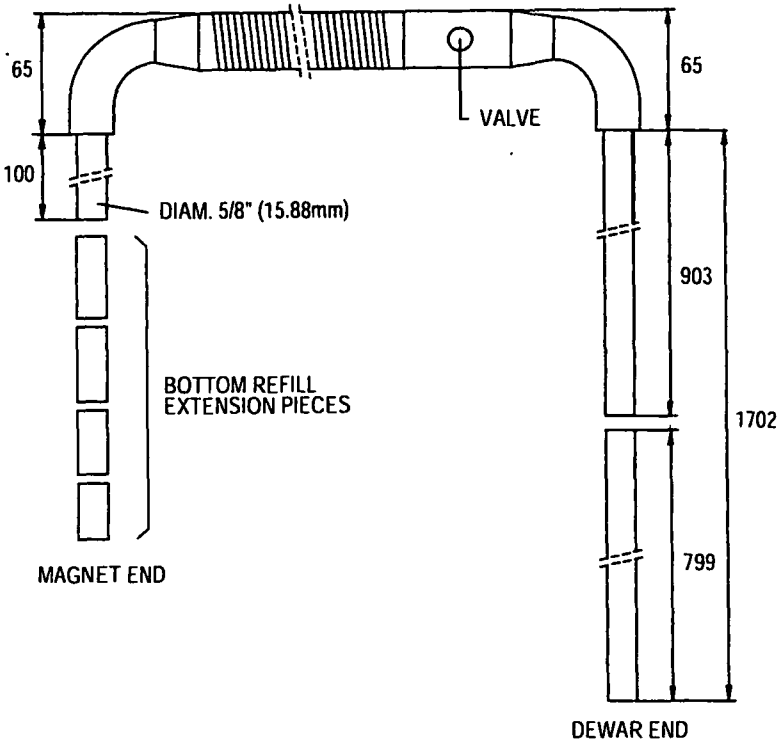


Figure 7 - IGC F2000 helium transfer line



Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

4.4. LIQUID HELIUM FILL PROCEDURE

Definition of the valves on a liquid helium dewar.

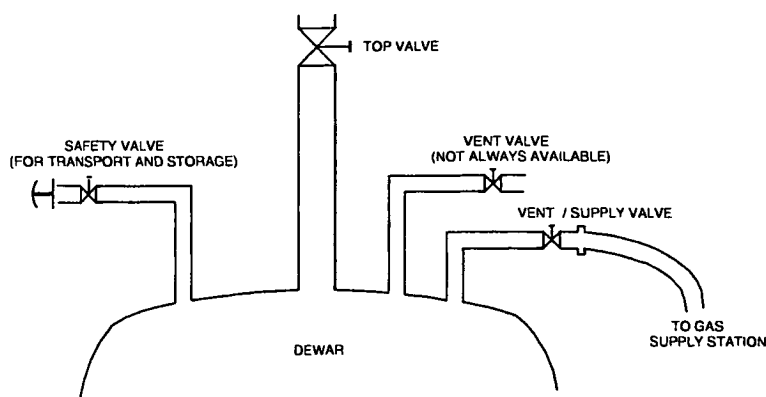
Safety valve	:	This is a valve that opens when the pressure inside the dewar gets higher than a certain pressure (e.g. 1 PSI). This valve MUST be open during transport and storage but closed during helium filling.
Vent/supply valve	:	This valve can be used to vent down the helium pressure inside the dewar. It is also used to supply external pressure from a helium gas supply station to the dewar in case of a non self-pressurizing dewar.
Vent valve	:	This valve can be used to vent down the helium pressure inside the dewar. This valve is not always available and not used in this procedure.
Top valve	:	This valve is on top of the dewar and is used to insert the transfer line.

NOTE

If the configuration of your dewar differs from this one, refer to the instructions on the dewar and/or call the supplier of the liquid helium for more information. If necessary, adapt the filling procedure.

In case self-pressurizing dewars are used for this helium fill, the vent/supply valve is then only used for venting, which simplifies the procedure

Figure 8 - Helium dewar valves configuration



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When the safety measures have been followed and the above items are on hand, fill the magnet using the following procedures:

1. Log the progress of the helium fill in the monitor chart of appendix A / Chapter 3.
2. When the IGC F2000 transfer line is **NOT** used; mark the magnet end of the helium transfer line such that it cannot be inserted deeper than 40 cm (16 inches) into the magnet. This to prevent a bottom fill.
3. Check if the magnet has sufficient over pressure by checking the helium pressure gauge reading. Pressure should be > 5 mBar.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Open the yellow ball valve.

Note that the yellow ball valve is ONLY open during helium filling and during ramping (up or down) of the magnet. It is closed at all other times. See Figure 9

4. Make sure that the (spare) compression fittings are in place at both ends of the transfer line. Check what transfer line length is needed for the dewar and, if necessary, attach an extender tube to the supply end of the transfer line.
 5. Move the helium dewar to a position where the transfer line can be inserted.
 6. Open the vent valve of the dewar and vent down the pressure to almost atmospheric pressure (i.e. flow nearly stops).
 7. Before inserting the transfer line into the dewar, provisionally close the exit on the magnet end of the transfer line with a rubber stop or place a rubber hose over the transfer line and close it by bending the hose.
 8. Open the valve on top of the dewar and insert the transfer line slowly into the dewar till it hits the bottom. If not yet done position the dewar in its final location. Raise the transfer line inside the dewar 2 cm so that eventual solid parts cannot block the entrance of the transfer line and tighten the compression fittings where the supply line enters the liquid helium dewar.
- Leave the vent valve of the dewar open.

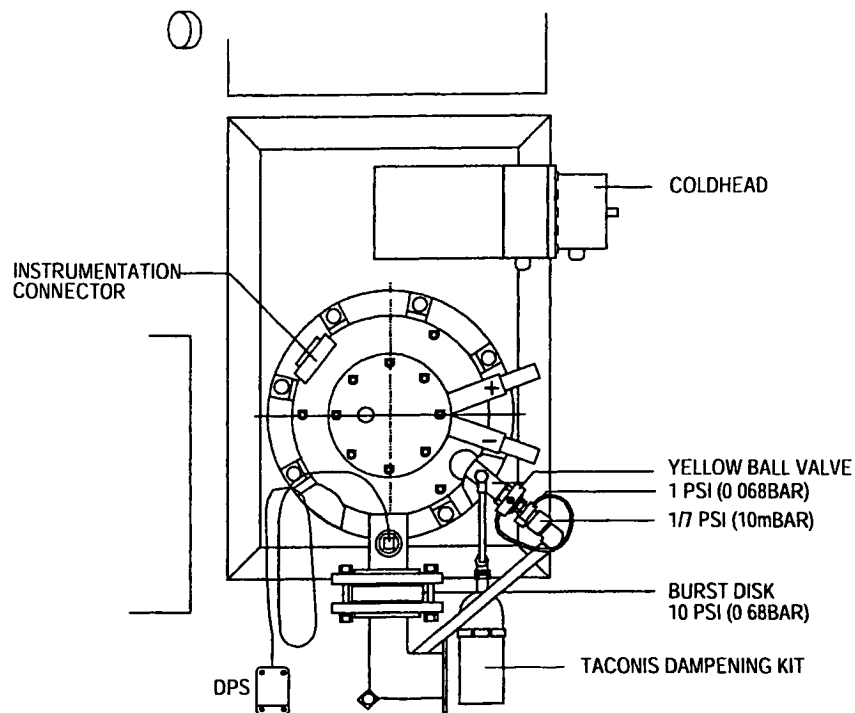
NOTE

With this procedure, frozen material may still block the line and slow down the transfer or even block the transfer entirely. If this happens, the line must be removed from the magnet and from the dewar and warmed up to room temperature before starting again.

-
9. Loosen the compression fitting of the helium fill port on top of the magnet.
 10. Close the safety vent valve on the dewar and, if present, remove the temporary stop on the magnet end of the transfer line.
 11. Externally pressurized dewar: Connect the helium gas supply station to the dewar via the vent/supply valve (do not forget to flush the air from the hose). Open the vent/supply valve on the dewar. Increase the pressure on the supply dewar, but keep the pressure below 140 mBar (2 psi).
 12. Self pressurizing dewars: Close the vent/supply valve of the dewar. Switch on the power of the pressure device of the dewar. Keep the pressure below 140 mBar (2 psi).
 13. Before inserting the transfer line into the magnet, it is essential that there is a 'slow and steady' stream of liquid helium passing through the transfer line. The associated heavy plume with flowing liquid helium has a distinctive appearance.
 14. Remove the compression fitting of the helium fill port and slowly insert the short end of the transfer line into the magnet. This insertion should take approximately 30 seconds.
 15. Once the magnet end of the transfer line is inserted 40 cm (16 inches)(or 10 cm for a F2000 type transfer line), tighten the associated compression fitting.
 16. Using the regulator and associated fittings and hoses, increase the pressure to about 350 mBar (about 5 psi) during transfer.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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Figure 9 - Top of the magnet



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17. Calculate the estimated helium fill level, while assuming an efficiency of 80%. Use the table in Appendix A / chapter 3 during the filling.
18. Monitor the progress of the helium transfer by taking a helium level reading every 5 minutes and estimate the time the transfer will be completed.
Helium level reading with a Magnet Monitoring Unit: Press and hold the helium read button for thirty seconds to display the level of liquid helium inside the helium vessel. The display will turn off when the button is released.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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Helium level reading with the F2000 magnet:

Using Gyroscan software:

- Login: GYROSCAN.
- Select: SCAN CONTROL.
- Select: SCAN UTILITIES.
- Select: READ HELIUM LEVEL.
Helium level is ...%

Using a DVM:

Connect a DVM, (set to 100 mV), to the voltage jacks on the RMMU
Press the HLI button.
Read the DVM after 30 seconds; reading in mV equals '%'.
Write down the value, date and time in the MRL.

CAUTION

When the supply dewar becomes empty, there is a chance to flush warm helium gas from the high pressure cylinder, through the supply dewar and into the magnet. This warm gas inside the magnet may quench the magnet or prevent a successful ramping up of the magnet. The magnet will quench during ramping up or shortly after being ramped UP. To avoid this, keep a close eye on the transfer, and listen for the 'whistling' sound associated with an empty supply dewar.

The transfer speed can vary between 250 and 500 litres/hour depending on the transfer line used. The time required depends on several factors, such as the supply pressure, length and size of the transfer line, the amount of liquid helium inside the supply dewar, and the backpressure of the valves. The maximum transfer pressure is 350 mBar (5 psi).

19. **When the dewar becomes empty**, one can hear a kind of 'whistling' sound coming from the transfer line. At this time the pressure inside the dewar will drop. **Open the safety valve on the dewar.**

The pressure on the dewar must be vented down to prevent possible transfer of warm helium gas to the magnet:

When externally pressurized dewars are used: Close the vent/supply valve. Close the pressure regulator of the helium supply station. Disconnect the gas supply station. Open the vent/supply valve and vent down the pressure in the dewar completely.

When self-pressurizing dewars are used: Switch off the self-pressurizing system on the dewar. Open the vent valve and vent down the pressure in the dewar completely.

NOTE

*Remove the compression fittings together with the transfer line.
Removing the transfer line through the fitting may cause damage to the O-ring in the fitting.*

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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20. When the pressure inside the dewar is (almost) vented down, remove the transfer line from the magnet and close the helium fill port of the magnet with its compression fitting.
21. Close the yellow ball valve on top of the magnet when no more dewars have to be transferred.
22. Move the dewar, with the transfer line still inserted, to a position where it is possible to remove the transfer line from the dewar.
23. Remove the transfer line from the dewar and close the valve on top of the dewar.
24. Open the safety valve and close the vent/supply valve of the dewar.
25. In case you want to transfer another (full) dewar into the magnet, heat up the transfer line to room temperature to prevent air from freezing solid inside the line. Record the amount of helium inside the magnet and the time of the reading. Return to point 4 of this procedure to continue with the helium transfer.
26. If not already done: **Close the yellow ball valve on top of the magnet.**
27. Record the HLI reading 1.5 hour after the transfer has been completed. It takes some time to establish a thermal equilibrium inside the helium vessel. Only then a reasonably accurate reading can be obtained.
28. Write the HLI reading and the amount of liquid helium used in the MRL.

4.5. TRANSFER PROBLEMS

In case the helium transfer has stopped (the helium level is not increasing in time) check the following items:

1. The pressure inside the helium vessel is equal to the pressure inside the dewar. Probably caused by a blocked venting system. Check if the yellow ball valve on top of the magnet is open.
2. The entrance to the transfer line inside the dewar is blocked by ice or because the transfer line hits the bottom of the dewar. In case the transfer line hits the bottom, raise it two centimetres and the transfer will continue.
3. The helium gas fill pressure has dropped because the helium gas cylinder is empty.

4.6. LEAK TEST

During the following actions you have to open the helium venting system of the magnet:

- Filling helium
- Ramping the magnet (up and down)
- Installation of the cold head

Every time when you have opened the magnet top you have to do a leak test on the helium vessel. If there is a leak inside the helium vessel, air might enter the vessel and freeze up solidly.

A leak test can be done if the pressure inside the magnet is high enough. In case there is no over pressure, create some extra boil off by switching 'ON' the B0-heater

Procedure:

1. Put SNOOP (delivered with the magnet spares) on the places where you have opened the magnet top.
2. When you see bubbles;
 - Tighten the connection or
 - Check the O-ring and replace if necessary

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
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5. APPENDIX A: MONITOR CHART HELIUM FILLING F2000 MAGNETS

Moment of fill Date: - - time: : h

Magnet type 0.5T / 1.0T / 1.5T

Equipment used

Transfer line used OMT / IGC-T5 / IGC-NT / Linde / Air products / IGC-F2000

Other

dewar contents[litres]

Last ramp

Date - -

at field no / yes

Starting point

Cold head running more than 5 days Yes / if no, for how many hours?h.

Connected to the gas-recovery no / yes

Liquid helium level before fill litre

Estimate fill level (eff. 80%) litre

Liquid helium level 1h. after fill litre

Efficiency %

Time	Helium level [litres]	actual pressure [mBar] [psi]	Fill pressure	remarks
00:00			< 140mB/2psi	insert Transfer line in dewar (≈30 seconds)
Before inserting the transfer line into the magnet, it is essential that there is a 'slow and steady' stream of liquid helium passing through the transfer line. The associated heavy plume with flowing liquid helium has a distinctive appearance.				
00:01			< 140mB/2psi	insert Transfer Line in magnet (≈30 seconds)
00:05			≈ 350mB/5psi	continue transferring at ≈350 mBar
00:15				
00:30				
00:45				
01:00				
00:15				
00:30				
00:45				
listen for the 'whistling' sound: stop transfer immediately and remove the pressure on the dewar				
02:00				
00:15				
03:00				Record helium level